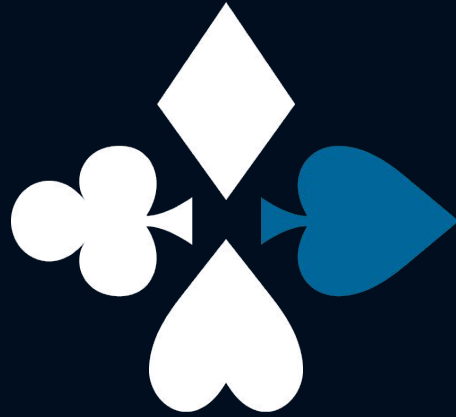




Pokerbots 2026

Lecture 3: Poker Strategy

Sponsors



CITADEL | CITADEL Securities





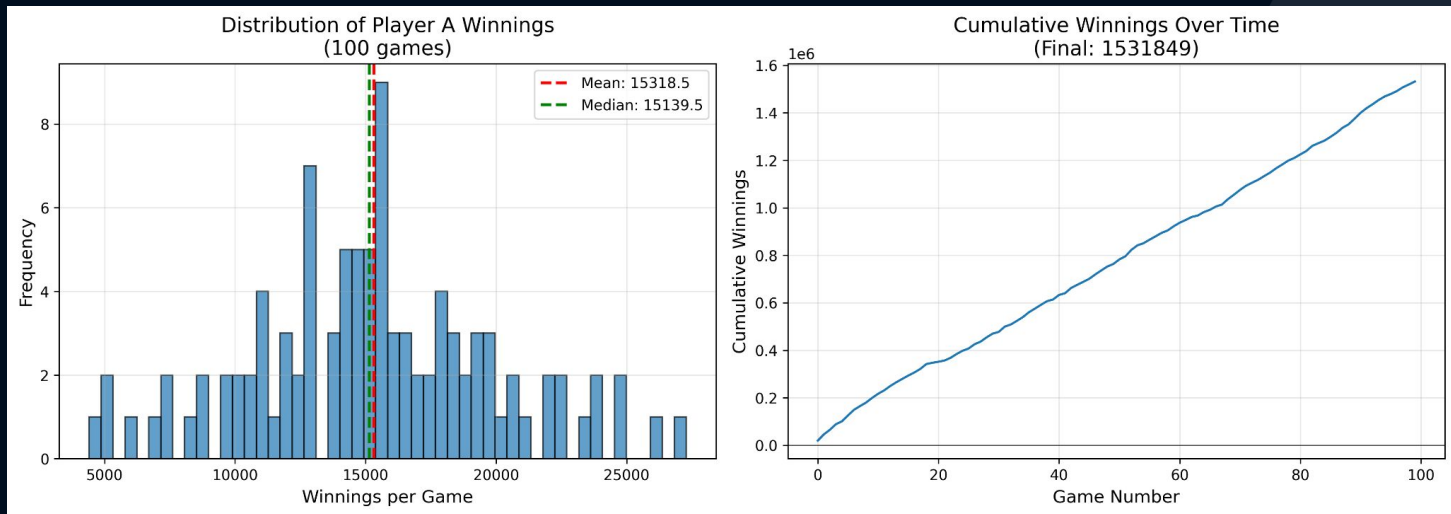
Announcements

Scrimmage Update!

- Scrimmage server running at pkr.bot/scrimmage
 - can't challenge people yet
 - we are working on it, hope to fix it as soon as possible
- First pokerbot submission due Friday 1/9, 11:59PM EST on scrimmage server
- Mini-tournament 1 will occur shortly after

Reference Bot 2 Posted

- See pkr.bot/refbot2
- Tosses smallest card
- Bets high when hole cards are high



Hackathon Next Week!

- Next Wednesday night (1/14)
- Show up and work on your pokerbot!
- Dinner provided, plan to go late into the night
- Snacks, fun, and games
- Prizes for challenges and those that stick around
- Location TBA

Giveaway Game:

pkr.bot/LUPI



- Enter an integer between 1 and 100, inclusive
- You win if you submit the lowest integer that nobody else also submits
- Submissions will close in 5 minutes



Poker Strategy and Theory

Sri Saraf

Agenda

- Basic Poker Principles
- Hand Types
- Pot Odds and Outs
- Implied Odds
- Board Texture Considerations
- Ranges



Poker Principles

Strength Principle

- Raising with Strong Hands
 - play aggressively for max value
- Checking with Middling Hands
 - want to get some chips but raising too strongly means you might lose when others call
- Folding/Bluffing Weak Hands
 - prevent giving your opponent too many chips
 - convincing your opponent to fold to avoid loss at showdown

Playing Styles

- Tight vs Loose
 - portion of hands played
- Aggressive vs Passive
 - betting and raising frequency
- In general, be tight and aggressive

Purpose of Betting

- Value Betting → want weaker hands to call
- Bluffing → want stronger hands to fold
- Betting for Protection → want drawing hands to draw at unfavorable odds
- Sometimes bets can serve multiple purposes

Deception

- Easy to read = Exploitable
- Almost never want to do one action at 100% frequency
- Incorporate bluffs



Hand Types

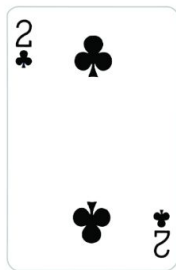
Starting Hand Types

- Paired vs Unpaired
- Suited vs Off-suit
- Connected vs Not connected
- High vs Low
- Best starting hands?

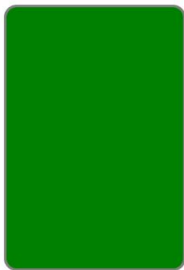
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Drawing Hand

Board Cards



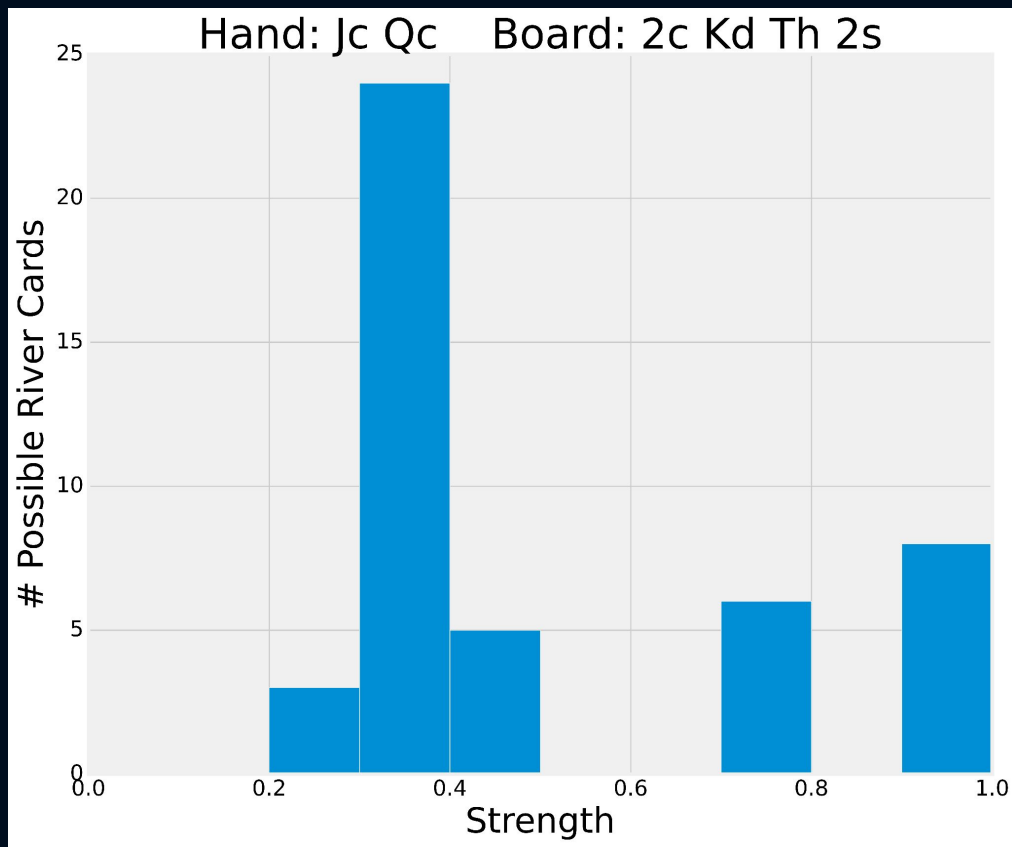
Opponent Cards



Your Cards



Drawing Hand

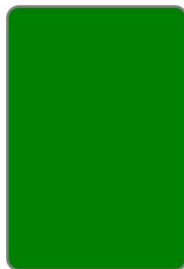
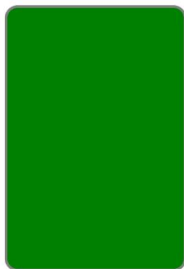


Low Pair

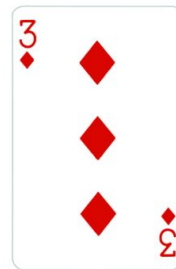
Board Cards



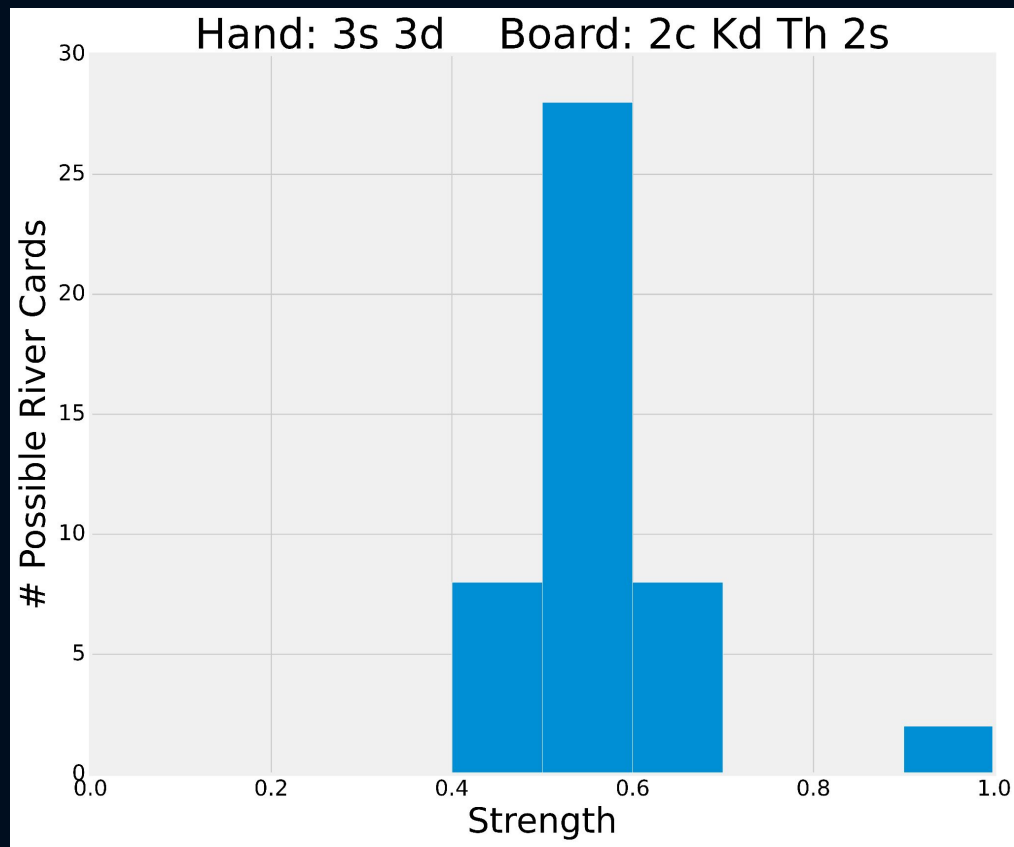
Opponent Cards



Your Cards



Low Pair

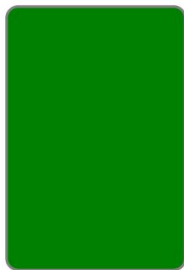
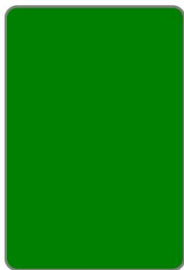


Top Pair

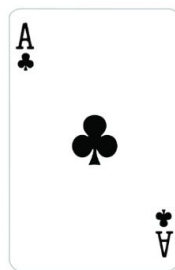
Board Cards



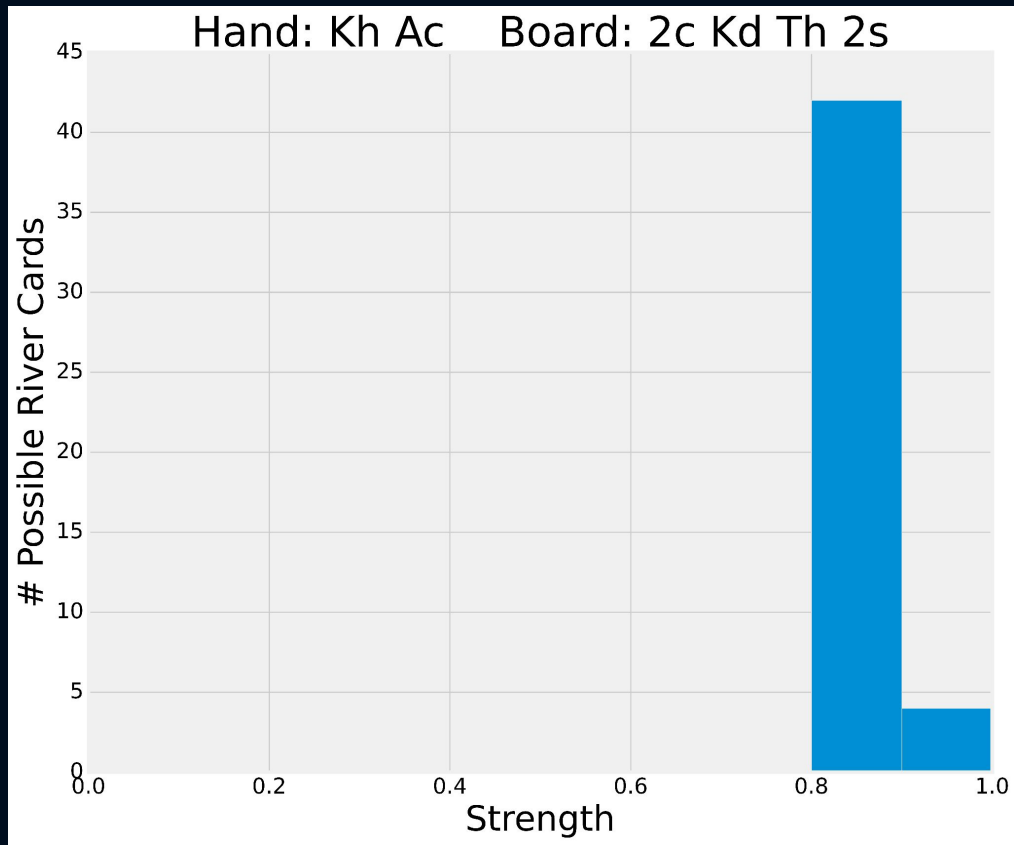
Opponent Cards



Your Cards

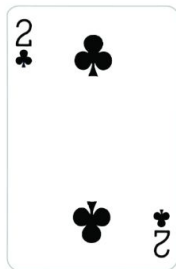


Top Pair

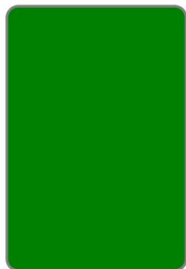
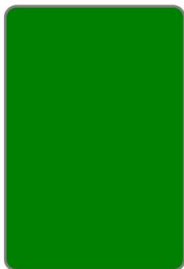


The Nuts

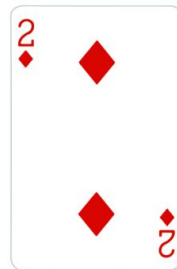
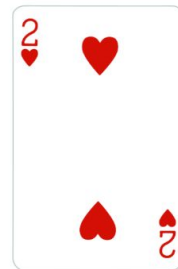
Board Cards



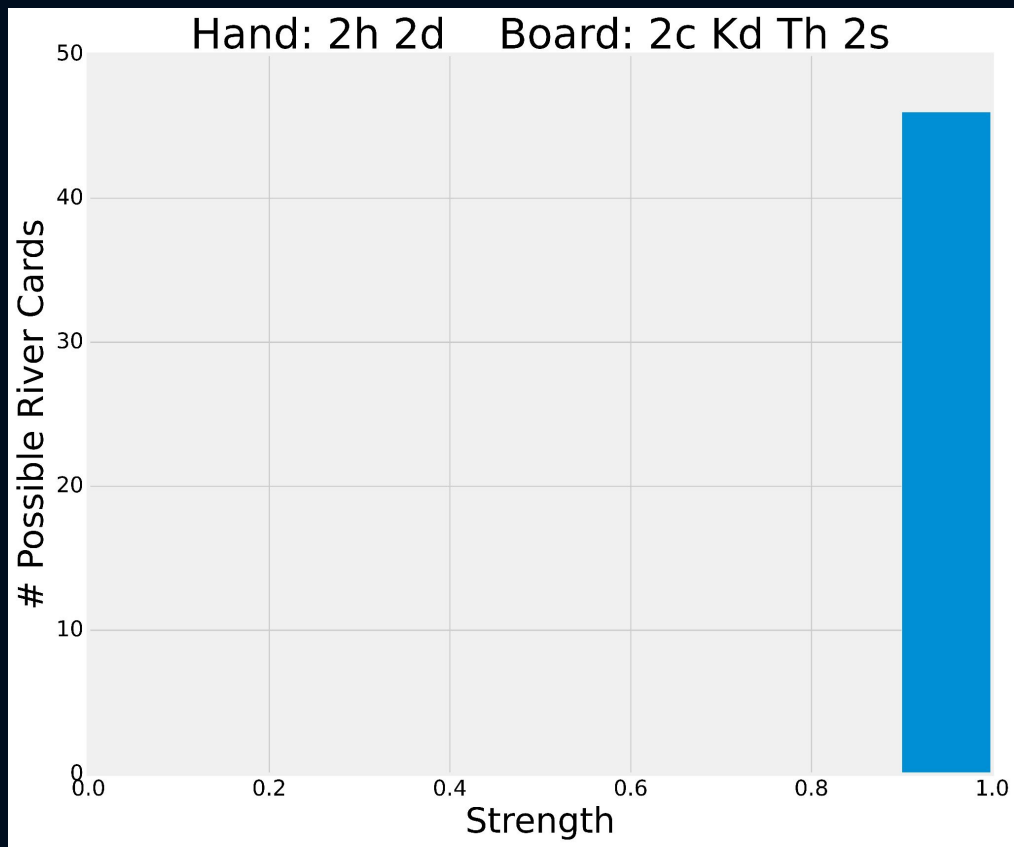
Opponent Cards



Your Cards



The Nuts

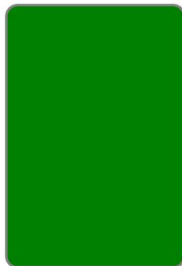
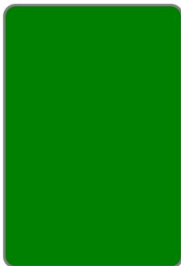


Total Air

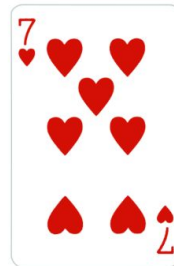
Board Cards



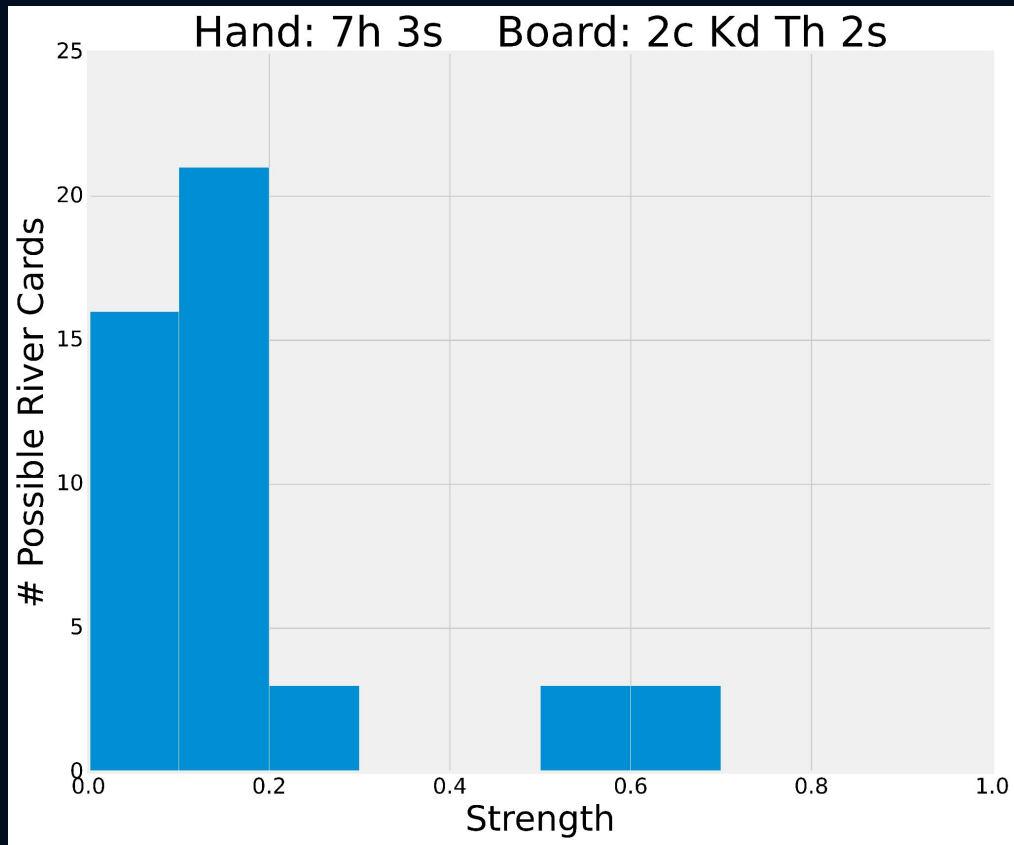
Opponent Cards



Your Cards



Total Air





Pot Odds and Outs

Defining P

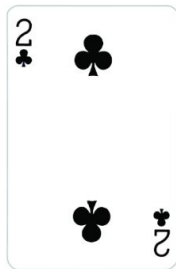
- At any point in the game there exists some fixed probability of winning (P)
 - Also called equity
- In reality P is often hard to explicitly calculate, but we can reasonably estimate

Expected Value

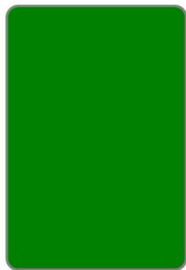
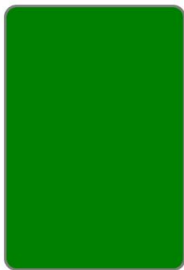
- Expected Value: how many chips we expect to win/lose on average
- EV of folding = $0 \cdot \text{pot_total} - 0 \cdot \text{continue_cost} = 0$
- EV of calling a bet = $P \cdot \text{pot_total} - (1 - P) \cdot \text{continue_cost}$
 - we want positive EV!!!
- $P \geq \text{continue_cost} / (\text{pot_total} + \text{continue_cost})$
- RHS is called the “pot odds”

Pot Odds Exercise

Board Cards



Opponent Cards



80

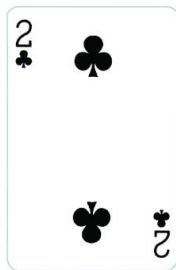


Your Cards

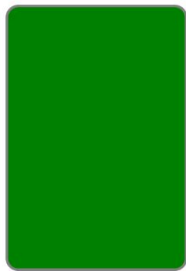
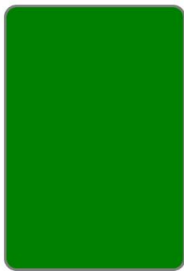


Pot Odds Exercise

Board Cards



Opponent Cards



10



80



Your Cards



C or F?

Pot Odds Exercise

- $\text{pot_total} = 90$
- $\text{continue_cost} = 10$
- $\text{pot odds} = 10 / (90 + 10) = 0.1$
- If $P \geq 0.1$, we should call!

How do we know what P is?

Estimating P by Counting Outs

Out = A card that would complete our hand or make us significantly stronger

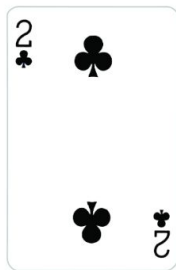
Idea: If we count our outs, we can estimate the probability of finding cards we need
We can use this to estimate our probability of winning P

Strategy:

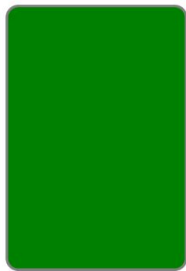
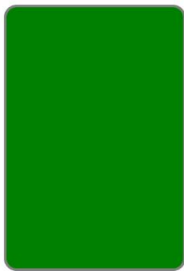
- Count the number of cards that complete our hand (outs)
- Multiply this number by 2 (1/52 cards gives ~2% chance of getting a specific card)
- If we have two cards left to see (turn and river), multiply by 2
- This number is our probability estimate! (as a percent)

Counting Outs Exercise

Board Cards



Opponent Cards



10



80



Your Cards



C or F?

Counting Outs

- How can we hit a straight?
 - Any Ace or Nine (8 outs)
 - Only 1 card left to come, so $1 * 8 * 2\% = 16\%$
- How can we hit second pair?
 - Any Jack or Queen (6 outs)
 - Only 1 card left to come, so $1 * 6 * 2\% = 12\%$
- P, assuming opponent does not have top pair or better: 28%
- $P > 0.1$, so we call!

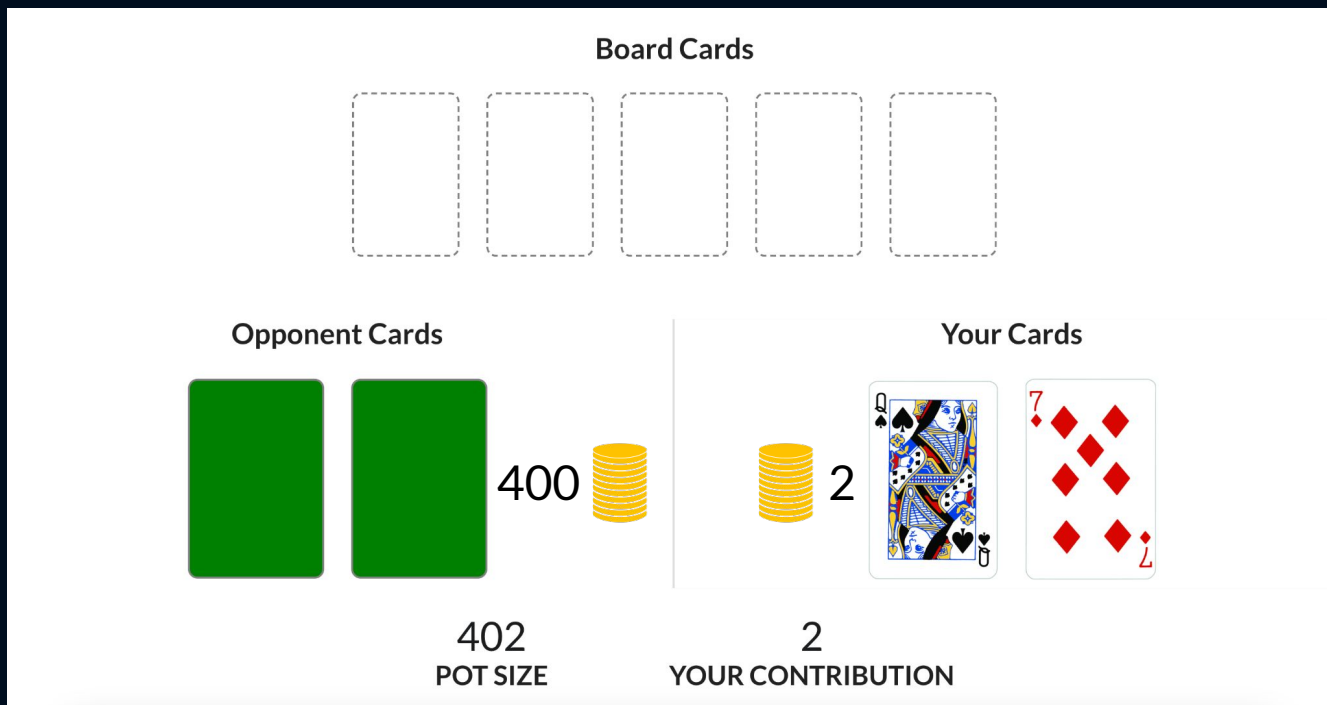
Did our opponent make a mistake?

Reverse Pot Odds

- If we overbet relative to the size of the pot, then we give our opponent the opportunity to exploit pot odds
- If they have a bad hand, we win a little
- If they have “the nuts,” we lose a lot

Example: All-in Bot

- Scenario: Opponent goes all-in before the flop every hand



Example: All-in Bot

- Scenario: Opponent goes all-in before the flop every hand
- What P do we need to call profitably as the big blind?
- $\text{pot_total} = 402$
- $\text{continue_cost} = 398$
- $\text{pot odds} = 398 / (402 + 398) = 0.4975$
- If $P \geq 0.4975$, we should call!
- Fun fact: Q5o tends to beat 50% of starting hands

Questions?





Implied Odds

Implied Odds

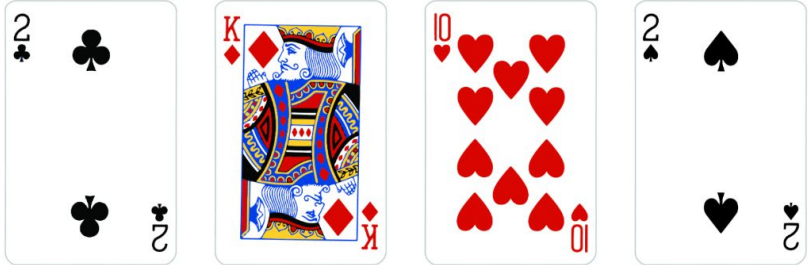
- The amount of money you expect to win on later streets if you hit one of your outs
- Enables us to call when we don't have the right pot odds
- Important to consider when we have a drawing hand

Updated P

- Pot odds cutoff: we should stay in the game if
$$P \geq \text{continue_cost} / (\text{pot_total} + \text{continue_cost})$$
- Implied odds cutoff: we should stay in the game if
$$P \geq \text{continue_cost} / (\text{pot_total} + \text{continue_cost} + \text{future_win_amount})$$

Pot Odds Exercise (again)

Board Cards



Opponent Cards



250 50

Your Cards



C or F?

300

200

The diagram illustrates a poker pot odds exercise. A central vertical line separates the 'Opponent Cards' side from the 'Your Cards' side. On the 'Opponent Cards' side, there are two green cards and a stack of 50 coins. On the 'Your Cards' side, there are two cards (Jack of Clubs and Queen of Clubs) and a stack of 300 coins. Above the cards, the 'Board Cards' are shown: 2 of Clubs, King of Diamonds, 10 of Hearts, and 2 of Spades. A stack of 200 coins is positioned above the central line. The text 'C or F?' is placed below the central line, likely referring to a call or fold decision.

Pot Odds Exercise (again)

- $\text{pot_total} = 250$
- $\text{continue_cost} = 50$
- $\text{pot odds} = 50 / (250 + 50) = 0.1667$
- $P, \text{ assuming you need to beat King pair} = 1 * 8 * 2\% = 0.16$
- $0.16 < 0.1667$, so we should fold!

Implied Odds

- Assumption, our opponent has three of a kind of Twos or pair of Kings and will call if we bet 100 more on the river no matter what card comes
- $\text{pot_total} = 250$
- $\text{continue_cost} = 50$
- $\text{future_win_amount} = 100$
- $\text{implied pot odds} = 50 / (250 + 50 + 100) = 0.125$
- $P, \text{ assuming you need to beat King pair} = 1 * 8 * 2\% = 0.16$
- $0.16 > 0.125$, so we should call!



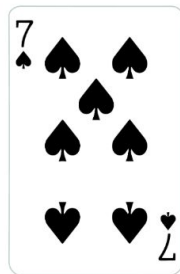
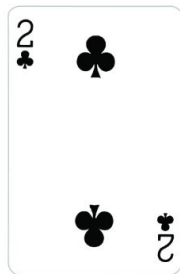
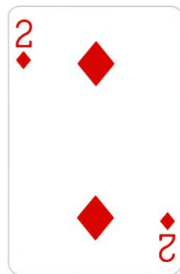
Board Texture

Board Texture

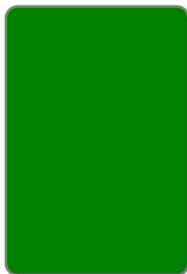
- How connected the board is
- Dry → hands are not likely to improve as more cards come
- Wet → hands are likely to improve as more cards come

Dry Board

Board Cards



Opponent Cards

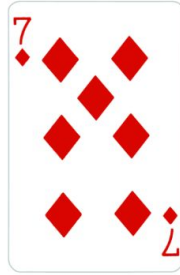


Your Cards

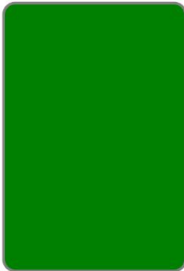


Wet Board

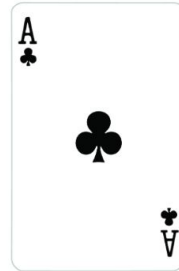
Board Cards



Opponent Cards

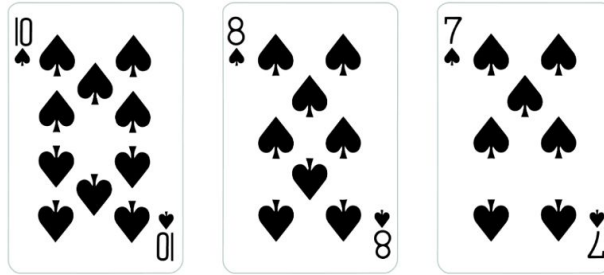


Your Cards



Very Wet Board

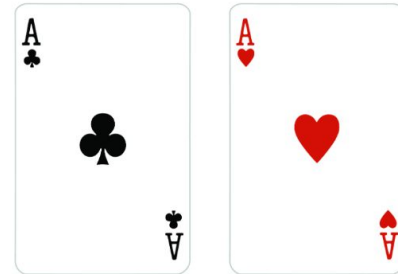
Board Cards



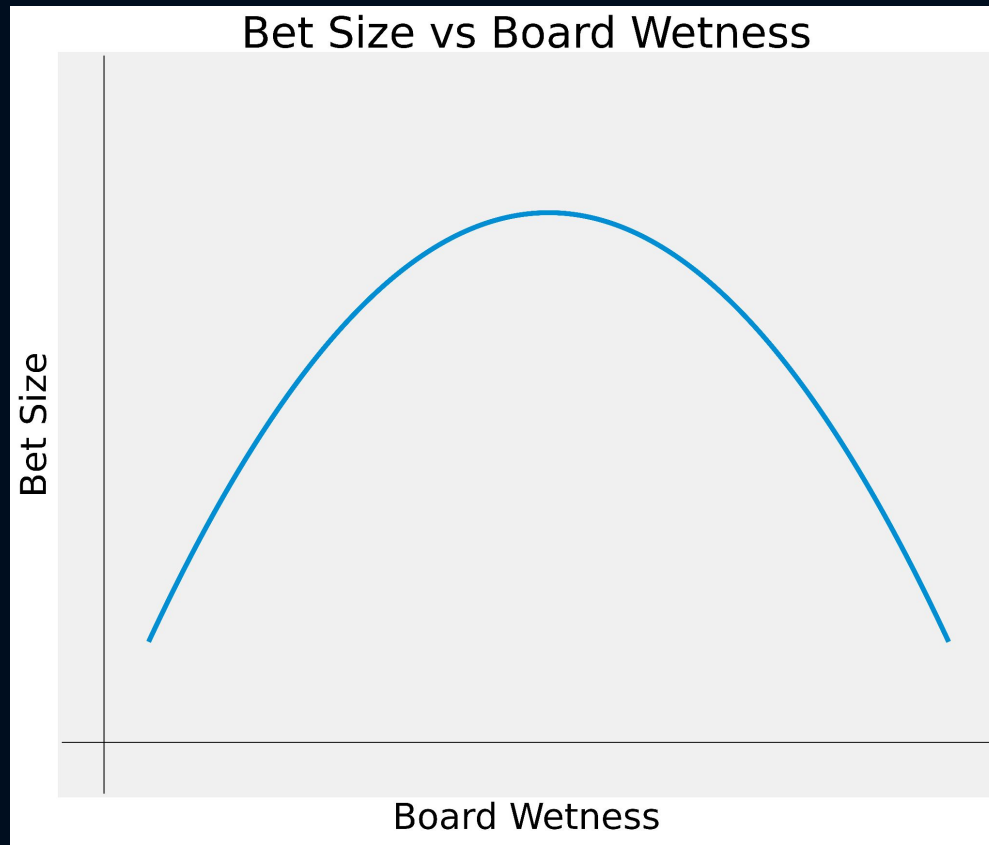
Opponent Cards



Your Cards

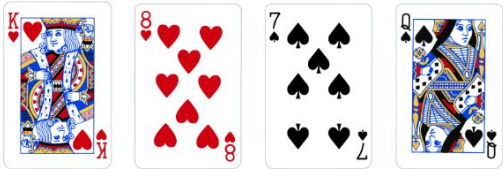


Flop Bet Size



Board Texture Exercise

Board Cards



Opponent Cards



350
POT SIZE

Your Cards



175
YOUR CONTRIBUTION

Choose an action

The cost is listed in parentheses

CHECK

RAISE (75)

RAISE (225)

Board Texture Exercise

- Two flush draws
- Several straight draws: 9T, TJ, AJ
- Board is wet, we don't want opponents to complete their draws cheaply
- Go all-in!



Ranges

Ranges

- We know the pot odds when faced with any bet
- If we can estimate P better than our opponent, then we will make money on average
- What affects P ?

Factors of win probability

- Bluffing
- Betting style
- Board and Hole cards
- *Ranges*

Our opponent's *range*
is the distribution of
hands we expect
them to hold

Which ranges are good?

- Tight-aggressive
- Fold early and often to mitigate losses
- Bet and win when you have a good hand!

Example range

- Six players, first to act
- Orange: Raise
- Blue: Fold

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Heads-Up Range

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
9.95	3.96	2.9	2.32	1.97	1.53	1.35	1.23	1.06	1.16	1.05	0.95	0.85
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
3.19	8.17	2.07	1.73	1.52	1.14	0.91	0.81	0.71	0.6	0.52	0.45	0.39
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
2.06	1.29	6.58	1.53	1.38	1.05	0.78	0.53	0.46	0.39	0.34	0.27	0.22
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
1.49	0.94	0.8	5.21	1.34	1.02	0.79	0.51	0.29	0.25	0.21	0.19	0.16
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
1.13	0.73	0.63	0.63	4.08	1.03	0.85	0.58	0.33	0.2	0.18	0.16	0.13
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
0.7	0.37	0.33	0.33	0.36	3.15	0.89	0.68	0.44	0.19	0.13	0.11	0.08
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
0.53	0.27	0.22	0.22	0.25	0.26	2.43	0.77	0.59	0.32	0.14	0.07	0.05
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
0.41	0.22	0.13	0.13	0.16	0.18	0.19	1.86	0.66	0.43	0.2	0.09	0.01
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
0.33	0.16	0.08	0.04	0.07	0.09	0.11	0.14	1.49	0.57	0.37	0.13	0.03
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
0.33	0.12	0.05	0.02	0	0.01	0.03	0.07	0.09	1.19	0.53	0.32	0.08
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
0.28	0.08	0.03	0	0	0	0	0	0.03	0.06	0.97	0.23	0.06
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
0.25	0.05	0	0	0	0	0	0	0	0	0	0.78	0.02
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22
0.2	0.02	0	0	0	0	0	0	0	0	0	0	0.61

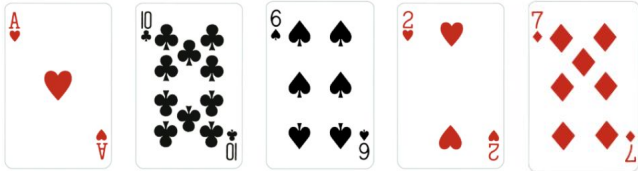
AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22



Variant Specific Considerations

Tossed Pot Odds

Board Cards



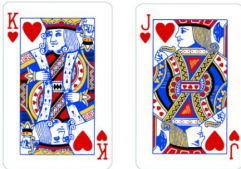
Opponent Cards



150 

C or F?

Your Cards



600
POT SIZE

225
YOUR CONTRIBUTION

Choose an action
The cost is listed in parentheses

CALL

FOLD (0)

Tossed Pot Odds

- EV of folding = $0 \cdot \text{pot_total} - 0 \cdot \text{continue_cost} = 0$
- EV of calling a bet = $P \cdot \text{pot_total} - (1 - P) \cdot \text{continue_cost}$
- $P \geq \text{continue_cost} / (\text{pot_total} + \text{continue_cost})$
- $P \geq 150 / (600 + 150) = 0.2$
- How many outs do we have?
- Without tossed card:
 - 4 Qs to the nuts: $2\% \cdot 4 = 8\%$
 - $0.08 < 0.2$, so we may not call!
- With tossed card:
 - 9 hearts left in the deck + 3 non-heart Qs to give us the nuts: $2\% \cdot 12 = 24\%$
 - $0.24 > 0.2$, so we can call!

Questions?



Lunch 🤪

Leave any type of feedback at pkr.bot/feedback!





Live Coding Session

But it's interactive?!?!?

pkr.bot/mc

Solutions can be found at

pkr.bot/mcsolutions

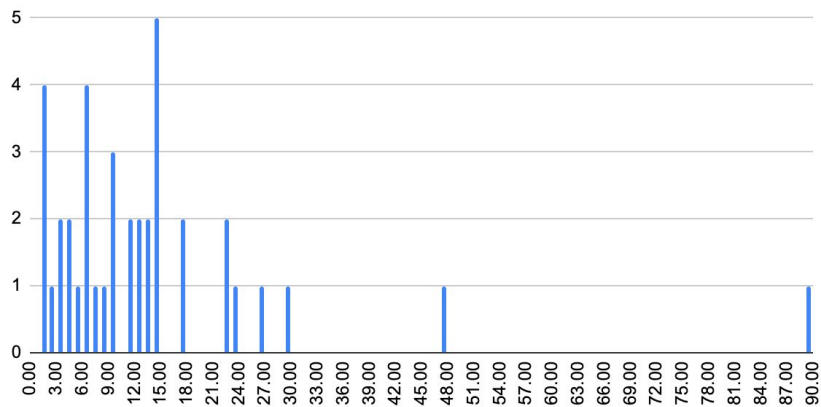
Giveaway Winners

LUPI Winner: *azjargal*

Lowest Unique Positive Integer was 2



Histogram of Submit an integer from 1 to 100



Submit an integer from 1 to 100

Thanks for watching!

Information hub at pkr.bot/hub

Slides/notes will be posted on pkr.bot/resources

Make sure to check pkr.bot/piazza for updates

Lecture recordings at pkr.bot/panopto

Leave feedback at pkr.bot/feedback!